

A Notation for sets

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ Set of nonnegative integers.
 $\mathbb{Z} = \{0, 1, -1, 2, -2, \dots\}$ Set of rational integers.
 $\mathbb{R}$ Real line.
 $\mathbb{R}_+ = \{x \in \mathbb{R} : x \geq 0\}$ Nonnegative real line.
 $\mathbb{R}_> = \{x \in \mathbb{R} : x > 0\}$ Strictly positive real line.
 $\mathbb{C}$ Complex plane.
 $\mathbb{C}^k$ (k th Cartesian power of $\mathbb{C}$) $= \mathbb{C} \times \mathbb{C} \times \dots \times \mathbb{C}$.
 $\mathbb{C}_> = \{z \in \mathbb{C} : z \neq 0, |\text{ph } z| < \pi/2\}$ Open right half-plane.
 $\mathbb{C}_0 = \{z \in \mathbb{C} : z \neq 0, |\text{ph } z| < \pi\}$ Complex plane cut along the nonpositive real line.
 $U = \{z \in \mathbb{C} : z \neq 0, -1, -2, \dots\}$ Complex plane punctured at the non-positive integers.
 $U_k = \{z = (z_1, \dots, z_k) \in \mathbb{C}^k : \sum_{i=1}^k z_i \neq 0, -1, -2, \dots\}$ Note $U_1 = U$.

B Integrals depending on a parameter

Special functions are often defined or represented by single or multiple integrals containing one or more parameters in the integrand; theorems about such integrals, considered as functions of the parameters, are indispensable. The theorems proved in textbooks usually make one or more simplifying assumptions: the integral is a single integral, or it is an integral with respect to Lebesgue measure or at most a positive measure, or the integrand contains only one parameter. (A notable exception is Fubini's theorem; integration under the integral sign is usually discussed in a much more general context than differentiation under the integral sign.) Even though the simplifying assumptions are commonly known to be inessential, it can be very difficult to find a theorem in print which strictly fits the case at hand. In this book the integrals often are multiple integrals with respect to a complex measure and depend on several parameters. On the other hand, there is nothing pathological or even unpleasant about the region of integration (usually $\mathbb{R}^k$, the unit simplex in $\mathbb{R}^k$, or a rectifiable Jordan curve in $\mathbb{C}$), the measure (usually Lebesgue measure, Dirichlet measure, or the line element in $\mathbb{C}$), or the integrand (usually continuous on the open region of integration). Stating theorems for measurable functions on a measure space may seem excessive generality, but this is a case where generality and simplicity go together.

The sets $\mathbb{N}$, $\mathbb{R}$, $\mathbb{C}$ are defined in Appendix A. If $X \subset \mathbb{R}^k$ is an open set, $C_n(X)$ is the class of functions $f : X \rightarrow \mathbb{C}$ which are continuous if $n = 0$

and n times continuously differentiable if $n \geq 1$. If $f \in C_n(X)$ and $m = (m_1, \dots, m_k) \in \mathbb{N}^k$, we write $\|m\| = \sum_{i=1}^k m_i$ and define, for every $x \in X$,

$$D_x^m f = \frac{\partial^{\|m\|} f}{\partial x_1^{m_1} \dots \partial x_k^{m_k}}, \quad 0 \leq \|m\| \leq n.$$

Since these derivatives are continuous by assumption, the order of differentiation is immaterial.

Associated with a complex measure μ defined on a measure space T is the total variation measure $|\mu|$ [see Rudin (1974, p. 125) or Hewitt (1965, p. 309)]. If μ is a positive measure, then $|\mu| = \mu$. The phrase "for almost every $t \in T$ " means everywhere in T except on a set of $|\mu|$ -measure zero. By $L_1(|\mu|)$ we mean the class of functions $f : T \rightarrow \mathbb{C}$ such that f is measurable and $\int_T |f| d|\mu| < \infty$.

To designate $f(x, t)$ considered as a function of t for fixed x , we shall write $t \mapsto f(x, t)$ or sometimes $[t \mapsto f(x, t)]$. Likewise the function $x \mapsto f(x, t)$ is the restriction of $f : X \times T \rightarrow \mathbb{C}$ to a fixed $t \in T$.

With the exception of Theorem B.5 the following theorems extend familiar properties of finite sums. For example a finite sum of analytic functions is analytic. Since integration is a form of addition, an integrand which is analytic in a parameter should have an analytic integral under suitable conditions. Indeed it does, and sufficient conditions are specified in Section B.4.

B.1 Lebesgue's theorem of dominated convergence

This fundamental theorem about taking a limit under the integral sign underlies Theorems B.3 and B.4. It is usually stated for a positive measure and a countable sequence of functions but can readily be extended to a function which depends continuously on a parameter [see Rudin (1974, pp. 27, 194)]. Moreover the measure can be complex since a complex measure is representable as a positive measure multiplied by a unimodular function [Rudin (1974, Section 6.12)].

THEOREM B.1 Let μ be a positive or complex measure on a measure space T , and let $|\mu|$ be the corresponding total variation measure. Let $Z \subset \mathbb{C}^k$ and let a be an accumulation point of Z . Assume that the function $f : Z \times T \rightarrow \mathbb{C}$ satisfies

- (i) $f_a(t) = \lim_{z \rightarrow a} f(z, t)$ exists for almost every $t \in T$,
- (ii) $[t \mapsto f(z, t)]$ is measurable for every fixed $z \in Z$,
- (iii) there exists a function $g \in L_1(|\mu|)$ such that $|f(z, t)| \leq g(t)$ for every $(z, t) \in Z \times T$.

Then $f_a \in L_1(|\mu|)$ and

$$\lim_{z \rightarrow a} \int_T f(z, t) d\mu(t) = \int_T f_a(t) d\mu(t).$$

Remark It is not required that $a \in Z$. For instance the usual statement of the theorem sets $k = 1$, $Z = \mathbb{N}$, and $a = \infty$.

B.2 Fubini's theorem

In the terminology used by Rudin (1974, Sections 1.18 and 6.7) a positive measure on a measure space T may take the value ∞ , but a complex measure satisfies $|\mu|(T) < \infty$. (Hence the positive measures are not a subclass of the complex ones.) In Fubini's theorem a positive measure, although not necessarily bounded, is required to be σ -finite [i.e., T is a union of countably many sets T_n such that $\mu(T_n) < \infty$]. For example Lebesgue measure on $\mathbb{R}$ is σ -finite but not bounded.

THEOREM B.2 Let λ be a positive σ -finite or complex measure on a measure space X , and let μ be a positive σ -finite or complex measure on a measure space T . Let $|\lambda|$ and $|\mu|$ be the corresponding total variation measures. Assume that the function $f : X \times T \rightarrow \mathbb{C}$ is measurable on the product space $X \times T$ (with respect to the product σ -algebra). If the integral

$$\int_X \int_T |f(x, t)| d|\lambda|(x) d|\mu|(t)$$

is finite when evaluated as an iterated integral in one order or the other, then the value of

$$\int_X \int_T f(x, t) d\lambda(x) d\mu(t)$$

is independent of whether it is evaluated as a double integral or an iterated integral in either order.

Remark For a more complete and precise statement of Fubini's theorem for positive measures, see Rudin (1974, Sections 7.8 and 7.9).

B.3 Continuously differentiable dependence on a real parameter

In the case that $T \subset \mathbb{R}^p$ and μ is Lebesgue measure, a proof and discussion of the following theorem are given by Garnir (1965).

THEOREM B.3 Let μ be a positive or complex measure on a measure space T , and let $|\mu|$ be the corresponding total variation measure. Let X be an open set in $\mathbb{R}^k$, and let $n \in \mathbb{N}$. Assume that the function $f : X \times T \rightarrow \mathbb{C}$ satisfies

- (i) $[x \mapsto f(x, t)] \in C_n(X)$ for almost every fixed $t \in T$,
- (ii) $[t \mapsto D_x^m f(x, t)] \in L_1(|\mu|)$ for every fixed $x \in X$ and every $m \in \mathbb{N}^k$ such that $0 \leq \|m\| \leq n$,
- (iii) for every compact set $K \subset X$ and every $m \in \mathbb{N}^k$ with $\|m\| = n$, there exists a function $g_K^m \in L_1(|\mu|)$ such that

$$|D_x^m f(x, t)| \leq g_K^m(t)$$

for every $(x, t) \in K \times T$.

Define

$$F(x) = \int_T f(x, t) d\mu(t)$$

for every $x \in X$. Then $F \in C_n(X)$ and

$$D_x^m F(x) = \int_T D_x^m f(x, t) d\mu(t)$$

for every $x \in X$ and every $m \in \mathbb{N}^k$ such that $0 \leq \|m\| \leq n$.

Remark If $n = 0$, the integral is continuous but not necessarily differentiable.

B.4 Analytic dependence on a complex parameter

The notation for partial derivatives is the same as before except that $x = (x_1, \dots, x_k)$ is replaced by $z = (z_1, \dots, z_k)$. A proof of the following theorem can be based on Theorem B.3 by setting $z = x + iy$ and applying the Cauchy–Riemann equations. Another proof uses Theorem B.2 in conjunction with Morera’s theorem.

THEOREM B.4 Let μ be a positive or complex measure on a measure space T , and let $|\mu|$ be the corresponding total variation measure. Let Z be an open set in $\mathbb{C}^k$. Assume that the function $f : Z \times T \rightarrow \mathbb{C}$ satisfies

- (i) $[z \mapsto f(z, t)]$ is holomorphic on Z for almost every fixed $t \in T$,
- (ii) $[t \mapsto f(z, t)] \in L_1(|\mu|)$ for every fixed $z \in Z$,
- (iii) for every compact set $K \subset Z$ there exists a function $g_K \in L_1(|\mu|)$ such that $|f(z, t)| \leq g_K(t)$ for every $(z, t) \in K \times T$.

Define

$$F(z) = \int_T f(z, t) d\mu(t)$$

for every $z \in Z$. Then F is holomorphic on Z and

$$D_z^m F(z) = \int_T D_z^m f(z, t) d\mu(t)$$

for every $z \in Z$ and every $m \in \mathbb{N}^k$.

B.5 Asymptotic dependence on a real parameter

The formula below, known as Laplace's approximation, is used in Section 3.4. For proofs see Pólya (1972, Part II, Number 201), Copson (1965, Section 18), and Olver (1974, Chapter 3, Section 7).

THEOREM B.5 Let I be an interval in $\mathbb{R}$ and let the functions $f : I \rightarrow \mathbb{R}_+$ and $\varphi : I \rightarrow \mathbb{C}$ satisfy the following conditions:

- (i) f attains its maximum at a point y in the interior of I , the supremum of f in any closed interval not containing y is strictly less than $f(y)$, and there is a neighborhood of y in which f'' exists and is continuous and strictly negative,
- (ii) φ is continuous at y , and $\varphi(y) \neq 0$,
- (iii) f and φ are Lebesgue measurable, and there exists $r \in \mathbb{R}$ such that $|\varphi| f^r$ is integrable on I .

Let $x \in \mathbb{R}$. Then, as $x \rightarrow +\infty$,

$$\int_I \varphi(t) [f(t)]^x dt \sim \varphi(y) [f(y)]^{x+1/2} \left[\frac{-2\pi}{xf''(y)} \right]^{1/2}.$$

B.6 Log-convex dependence on a real parameter

Since a finite sum of log-convex functions is log-convex (Exercise 3.5-1), an integrand which is log-convex in a parameter is expected to have a log-convex integral. Usually the sum or integral is strictly log-convex even if the summand or integrand is not.

THEOREM B.6 Let μ be a positive measure on a measure space T , and assume that $\mu(T) \neq 0$ and μ is not concentrated on one point in T . Let I be an interval in $\mathbb{R}$. Assume that the function $f : I \times T \rightarrow \mathbb{R}_>$ satisfies

- (i) $[x \mapsto f(x, t)]$ is log-convex for almost every fixed $t \in T$,
- (ii) $[t \mapsto f(x, t)] \in L_1(\mu)$ for every fixed $x \in I$,
- (iii) if $[t \mapsto f(x, t)/f(y, t)]$ is constant almost everywhere on T , then $x = y$.

Define

$$F(x) = \int_T f(x, t) d\mu(t)$$

for every $x \in I$. Then F is strictly log-convex on I .

Proof Since f is strictly positive, so is F . Let $x, y \in I$ and $u \in \mathbb{R}$, and assume $x \neq y$ and $0 < u < 1$. By assumption (i) and (3.5-3),

$$\begin{aligned} F[ux + (1-u)y] &= \int_T f[ux + (1-u)y, t] d\mu(t) \\ &\leq \int_T [f(x, t)]^u [f(y, t)]^{1-u} d\mu(t) \\ &< \left[\int_T f(x, t) d\mu(t) \right]^u \left[\int_T f(y, t) d\mu(t) \right]^{1-u}. \end{aligned}$$

The last inequality is Hölder's inequality for integrals [see Hardy (1952, Section 6.9)], and the inequality is strict because assumption (iii) implies that $f(x, t)$ is not proportional to $f(y, t)$. Thus

$$F[ux + (1-u)y] < [F(x)]^u [F(y)]^{1-u},$$

whence F is strictly log-convex. $\square$

Solutions to Exercises

Chapter 2

2.2-1 Use (2.2-10) and note that $(|a| + k)/|a| \leq (r + k)/r$, $k \in \mathbb{N}$.

2.2-2 $(2, n)/(1, n)$; $(\frac{3}{2}, n)/(\frac{1}{2}, n)$; $(\frac{5}{4}, n)/(\frac{3}{4}, n)$; $-(\frac{1}{2}, n)/(-\frac{1}{2}, n)$; $-(\frac{3}{2}, n)/(-\frac{1}{2}, n)$; $(a + 1, n)/(a, n)$; $(a + 2, n)/(a, n)$.

2.2-3 $(2m)!! = 2^m(1, m)$; $(2m + 1)!! = 2^m(\frac{3}{2}, m)$; $(2m - 1)!! = 2^m(\frac{1}{2}, m)$.

2.2-4 Apply (2.2-5) to both sides of $(a, m - n) = (a, -n + m)$ and use (2.2-6). Each member has the value $\frac{1}{2}$.

2.2-5 Multiply both sides by $(1, n)$ and use (2.2-7). If $n = -N < 0$, $(1, n)$ is not defined, but $(1 - N, -N) = (-1)^N/(N, N) = (-1)^N(N - 1)! 2N/(1, 2N)$. Apply (2.2-7) to the last factor and then (2.2-6) to get $(1 + n, n) = 2 \cdot 4^n(\frac{1}{2}, n)$, $n = -1, -2, \dots$.

2.2-6 (i) Well defined only if $n \geq 0$. Use (2.2-2).

Parts (ii)–(v) also can be proved using (2.2-2) directly, but it becomes tedious if n , k , and $n - k$ can be positive or negative.

(ii) $(a, n) = (a, n - 1)[(a - 1) + n]$; $(a, n) = [(a + n) - n](a + 1, n - 1)$.

(iii) Multiply and divide the left side by $(a, -n)$ and use (2.2-5) and (2.2-6).

(iv) Multiply the left side by (a, k) and divide by $(a, n - n + k) = (a, n)(a + n, -n + k)$.

(v) Multiply and divide the left side by (a, n) .

(vi) Same as (iii) with a replaced by $a - n$.

(vii) Multiply and divide the left side by $(\frac{1}{2} + a, -n)$ to show that it equals $(-1)^n(\frac{1}{2} + a, n)(\frac{1}{2} - a, n)$ and so is even in a .

2.2-7 Use (2.2-8) and (2.2-9).

2.2-8 Proofs are similar to the proof of (2.2-7).

2.2-9 Use $|a + k| \geq |\operatorname{Re}(a + k)| = |\operatorname{Re} a + k|$, $k = 0, 1, 2, \dots, n - 1$.

2.3-1 Use (2.2-9), (2.3-1), and Exercise 2.2-6(iii).

2.3-2 Use (2.2-12) and (2.3-1).

2.3-3 Use Exercise 2.2-7, (2.3-1), and Exercise 2.2-6(vi).

2.3-4 Set $b = 1$ in (2.3-1). The sum is $(a + 1, n)/n!$.

2.3-5 Set $p = q = n$ in Exercise 2.3-2. For the second equation use (2.2-8), (2.2-9), (2.3-1), and (2.2-12).

2.3-6 Note that $(-1, m) = (-1, m)/m!$, $m \in \mathbb{N}$, and use (2.3-1). The sum is $(-k, n)/n!$.

2.3-7 Show that $\binom{a+m}{s} = (-1)^{s+m}(1+a, m)(-a, s-n)(-a+s-n, n-m)/s!$ and use (2.3-1). The sum is $\binom{a}{s-n}$ and vanishes if $n > s$.

2.3-8 Show that $(q+m+n)!/(q+n)! = (-1)^{p-n}(1+q+m, n)(-q-p, p-n)(q+m)!/(q+p)!$ and carry out the sum over n by (2.3-1). In the remaining summation only the term with $m = p$ fails to vanish, and the sum is $1/p!(q+p)!$.

2.3-9 Use (2.2-12), note that $(1-a-m, m) = (-1)^m(a, m)$, and use (2.3-1). The sum is δ_{n0} .

2.3-10 Use (2.2-12), show that $(1-a-m, n-1) = (-1)^{m-1}(a, m)(-a, n-m)/a$ if $a \neq 0$, and use (2.3-1). If $a = 0$, use continuity since the sum is a polynomial in a .

2.3-11 The term with $m = 0$ vanishes. Set $m = s+1$, $n = N+1$, $(a, s+1) = a(a+1, s)$, and use (2.3-1). The sum is $a(a+b+1, n-1)/(n-1)!$.

2.3-12 Set $r = s+1$, $n = N+1$, and note that $(1+s, p)/p! = (1+p, s)/s!$ and $(N+1-s, q)/q! = (1+q, N-s)/(N-s)!$. Use (2.3-1).

2.3-13 The sum over n can be done with the help of (2.3-1) if $a \in U$ to yield $(-1)^N \binom{M}{N}(a, M)/(a, N)$, which is 0 if $M < N$. Interchange M with N and a with b to obtain the sum over m . Hence the double sum is δ_{MN} (for all a, b by continuity).

2.4-1 Multiply the two series and use Exercise 2.3-3. Set $a = b = 1$ and use (2.4-15).

2.4-2 See Exercise 2.3-1.

2.4-3 Write ${}_2F_0(-n, \alpha; z)$ as a sum with summation index m , choose $n-m$ as a new summation index, and use (2.2-8). The relations for Hermite and Laguerre polynomials are corollaries.

2.4-4 Proceed as in Exercise 2.4-3.

2.4-5 Put $m = n+s$ and use (2.2-5).

2.4-6 Substitute (2.4-9), differentiate term by term using $\delta z^n = nz^n$, and change the summation index on one side by unity.

2.4-7 Differentiate term by term, change the summation index, and use (2.2-5).

2.4-8 Separate ${}_0F_1(c; z)$ into even and odd terms and use (2.2-7). For the Kelvin functions set $c = 1$ and use (2.4-15).

2.4-9 Set $y = xz$, multiply the two series, and collect coefficients of x^n . Use (2.2-5) and (2.2-6).

2.4-10 Let $r = \max\{|x|, |y|\}$. Use (2.2-10) and (2.3-1) to show that (2.4-25) is majorized by $\sum_{N=0}^{\infty} |(a, N)/(c, N)|(|b| + |b'|, N)r^N/N!$, which converges by the ratio test.

2.5-1 The product is the limit as $n \rightarrow \infty$ of $(\frac{1}{2}, n)(\frac{3}{2}, n)/(n!)^2$. Use (2.5-2).

2.5-2 Apply (2.2-7) to $(2n)! = (1, 2n)$ and use (2.5-3).

2.5-3 The product is the limit as $n \rightarrow \infty$ of $(\frac{1}{2}, n)^2/(\frac{1}{4}, n)(\frac{3}{4}, n) = [(\frac{1}{2}, n)/n!][(2n)!/(\frac{1}{2}, 2n)]$ by (2.2-7). Apply (2.5-3) to each quantity in brackets.

2.5-4 By (2.5-7), $(\pi n)^{-1/2}n!/(\frac{1}{2}, n) = (1 + 1/2n)^{1/2}(1 + \xi/2n)^{-1/2}$, where $0 < \xi < 1$. The right side is greater than 1 and does not exceed $(1 + 1/2n)^{1/2} < 1 + 1/4n$.

2.6-1 For every positive $t \neq 1$, $\log t = \int_1^t du/u < \int_1^t du$ and hence $\log t < t - 1$. If $t_i = x_i/\sum u_j x_j$, $i = 1, \dots, k$, then $t_i \neq 1$ for some value of i (except in the trivial case when all the x_i are equal). Hence $\sum u_i \log t_i < \sum u_i(t_i - 1) = 0$, which implies $\sum u_i \log x_i < \log \sum u_j x_j$.

2.6-2 In the first member of the inequality following (2.6-4), replace $z^2/3$ by $z^2/(3 - z^2)$.

2.6-3 Choose $x = e^{i\theta}$, $y = e^{-i\theta}$, where $0 < \theta \leq \frac{1}{2}\pi$. Note that the conditions imposed on x and y in Theorem 2.6-3 are satisfied if $0 < \theta < \pi/4$.

2.6-4 The terms on the left are $\frac{1}{2}(x^{1/2} \pm y^{1/2})(\bar{x}^{1/2} \pm \bar{y}^{1/2})$. Expand and add the products with upper and lower signs. Since $2a = |x| + |y|$ and the interfocal distance is $2c = |x - y|$, the minor axis is $2b = 2(a^2 - c^2)^{1/2} = |(x\bar{y})^{1/2} + (\bar{x}y)^{1/2}|$.

2.6-5 Set $x = e^{i\theta}$, $y = e^{-i\theta}$ in Exercise 2.6-4.

2.6-6 Draw the triangle with vertices 0, x , y and the bisector of the angle (say 2θ) at 0. Let d be the distance from 0 to the midpoint of the opposite side, which the bisector meets at angle φ . Both values of $(xy)^{1/2}$ lie on the bisector, and $d^2/|xy| = \sin(\varphi + \theta)\sin(\varphi - \theta)/\sin^2 \varphi = 1 - \sin^2 \theta/\sin^2 \varphi < 1$.

2.7-1 By Exercise 2.6-2 the first member of (2.7-6) can be replaced by $\sum_{k=n}^{2n-1} 1/(k^2 + k + \frac{1}{2}) \geq \sum_{k=n}^{2n-1} [(k+x)^{-1} - (k+1+x)^{-1}] = n/(n+x)(2n+x) > (1/2n)(1 + 1/12n^2)^{-1}(1 + 1/24n^2)^{-1} > (1/2n)(1 - 1/8n^2)$, where $x = \frac{1}{2}(2n+1)^{-1}$. Hence $s = (1/2n) - \theta/16n^2$.

2.7-2 By Exercise 2.7-1 and (2.5-7), $\log[n!(2\pi)^{-1/2}n^{-n-1/2}e^n]$ is greater than $-1/24n$ and less than $\frac{1}{2}\log(1 + 1/2n) - (1/24n) + (1/192n^3)$. Expanded in powers of $1/n$, this quantity is an alternating series of decreasing terms and so does not exceed its first term, which is $5/24n$.

Chapter 3

3.2-1 If $z = 0$ the integral diverges logarithmically. If $z = x + iy \neq 0$ and $0 < \delta < 1$, then $\int_{\delta}^1 t^{z-1} dt = z^{-1}(1 - \delta^z) = z^{-1}[1 - \delta^x \exp(iy \log \delta)]$. As $\delta \rightarrow 0$, this tends to z^{-1} if $x > 0$, oscillates if $x = 0$ and $y \neq 0$, and becomes infinite if $x < 0$.

3.2-2 Apply Cauchy's theorem to the integral around a contour consisting of segments of γ and the positive real line joined by arcs of circles with center 0 and radii r and R . Show that the integrals along the circular arcs tend to 0 as $r \rightarrow 0$ and $R \rightarrow \infty$.

3.2-3 Let $t = s/z$ and use Exercise 3.2-2.

3.2-4 Set $at^v = s$ and use Exercise 3.2-2.

3.2-5 Use Exercise 3.2-3. If $p = q = 0$, the series in the integrand is e^{xt} and the integral diverges if $\operatorname{Re}(x - s) > 0$.

3.3-1 If $z \in \mathbb{C}_+$, split (3.24) into two parts. The Taylor series of e^{-t} converges uniformly for $0 \leq t \leq r$ and can be integrated term by term. The integrated series converges uniformly in z on compact subsets of U , and its sum is analytic on U . Since the remaining integral is an entire function of z (see Appendix B.4), the asserted equation holds on U by the permanence of functional relations.

3.4-1 In (3.4-3) evaluate $g(x)$ by using Theorem B.5 with $f(t) = te^{-t}$ and $y = 1$.

3.4-2 Use Exercise 3.4-1 and note that $(1 + a/x)^{x+a-1/2} \rightarrow e^a$ as $x \rightarrow \infty$.

3.4-3 If $m - 1 \in \mathbb{N}$, then $(1, n)n^{m-1}/(m, n) = 1 \cdot 2 \cdots (m-1)n^{m-1}/(n+1)(n+2) \cdots (n+m-1) \rightarrow (m-1)! = \Gamma(m)$ as $n \rightarrow \infty$.

3.4-4 Use Exercise 3.2-3, (3.2-9), and Corollary 3.4-3.

3.4-5 Use (3.4-1) and (3.4-5); note that $n! = (1, n)$.

3.4-6 Use (3.4-5) after writing the left side as $\lim(a_1, N) \cdots (a_k, N)/(c_1, N) \cdots (c_k, N)$.

3.5-1 (i) Convexity of $f + g$ follows trivially from the definition. If $f(x) = g(x) = x$, then fg is convex, but if $f(x) = x^2$ and $g(x) = e^{-x}$, fg is not convex on $\mathbb{R}_+$.

(ii) Log-convexity of fg follows trivially from the definition. Let $a = f(x)$, $b = f(y)$, $c = g(x)$, $d = g(y)$. Since $f[ux + (1-u)y] \leq a^u b^{1-u}$ (and similarly for g), $f + g$ is log-convex if $a^u b^{1-u} + c^u d^{1-u} \leq (a+c)^u (b+d)^{1-u}$. The ratio of the left side to the right is $\alpha^u \beta^{1-u} + (1-\alpha)^u (1-\beta)^{1-u}$, where $\alpha = a/(a+c)$ and $\beta = b/(b+d)$. By Exercise 2.6-1 this does not exceed $u\alpha + (1-u)\beta + u(1-\alpha) + (1-u)(1-\beta) = 1$.

3.5-2 By strict log-convexity $\Gamma(\frac{3}{2}) < [\Gamma(\frac{1}{2})\Gamma(1)]^{1/2} = \pi^{1/4}$.

3.5-3 The lower bound is derived by combining $\Gamma(\frac{1}{2}) < [\Gamma(\frac{1}{2})\Gamma(\frac{3}{2})]^{1/2}$ with Exercise 3.5-2. For the upper bound, $(\frac{1}{2})\Gamma(\frac{1}{2}) = \Gamma(\frac{1}{2}) < [\Gamma(1)\Gamma(\frac{3}{2})]^{1/2} = [\frac{1}{2}\Gamma(\frac{1}{2})]^{1/2}$.

3.5-4 If $y = x + n$, so that $0 < y < 1$, then $(-1)^n \Gamma(x) = \Gamma(y)(1-y)^{-1} \cdots (n-y)^{-1}$. This product of strictly log-convex factors is strictly log-convex by Exercise 3.5-1.

3.5-5 By (2.2-12) and (2.2-6), $\binom{x}{m} = [(1+x, -m)\Gamma(m+1)]^{-1}$. Use (3.2-9) and Theorem 3.5-3.

3.5-6 Since the divided differences are symmetric in their arguments, we may assume $x > y > z$. Then f is monotonically increasing if and only if $f(x) \geq f(y)$ for all such x and y , which is equivalent to (i). Also, f is convex if and only if $uf(x) - f[ux + (1-u)z] + (1-u)f(z) \geq 0$ for $0 < u < 1$ and for all $x, z \in I$ with $x > z$. Define $y = ux + (1-u)z$ to show that this is equivalent to (ii).

3.5-7 If $k = 2$, the inequality holds by definition of convexity. If $k > 2$, make the inductive hypothesis that it holds for $k-1$. Since the case $u_k = 1$ is trivial, assume $u_k < 1$ and define the convex combination $X = \sum_{i=1}^{k-1} u_i x_i / (1 - u_k)$. Then $f(u \cdot x) = f[(1 - u_k)X + u_k x_k] \leq (1 - u_k) \times f(X) + u_k f(x_k) \leq \sum_{i=1}^{k-1} u_i f(x_i) + u_k f(x_k)$.

3.5-8 To prove sufficiency note that there exists i such that $u_i > 0$. Then, for every $j \neq i$, either $x_j = x_i$ or $u_j = 0$. Hence $u \cdot x = x_i$ and equality holds. To prove necessity assume there exist i and j such that $(x_i - x_j)u_i u_j \neq 0$. Let $w = u_i / (u_i + u_j)$, $a = u \cdot x + u_j(x_i - x_j)$, and $b = u \cdot x + u_i(x_j - x_i)$. Then $wa + (1-w)b = u \cdot x$ and $(a-b)w(1-w) \neq 0$, whence $f(u \cdot x) < wf(a) + (1-w)f(b)$. Now $f(a) \leq \sum_{m=1}^k u_m f(x_m) + u_j[f(x_i) - f(x_j)]$ and $f(b) \leq \sum_{m=1}^k u_m f(x_m) + u_i[f(x_j) - f(x_i)]$. Hence $f(u \cdot x) < \sum_{m=1}^k u_m f(x_m)$.

3.6-1 Note that $\sum_{k=1}^n \log(1 + 1/k) = \log(n+1) = \log n + \log(1 + 1/n)$.

3.6-2 By (3.6-7) and (3.6-11), $0 < \gamma < \frac{1}{2} \sum_{n=1}^{\infty} n^{-2} < \frac{1}{2} + \frac{1}{2} \sum_{n=2}^{\infty} [(n-1)^{-1} - n^{-1}] = 1$. Using $\sum_{n=1}^{\infty} n^{-2} = \pi^2/6$, we obtain $0 < \gamma < \pi^2/12$.

3.6-3 By (2.6-5), $1 < (n + \frac{1}{2}) \log(1 + 1/n) < 1 + (1/12n) - 1/12(n+1)$. By (3.6-11), $(5A + B)/6 < \gamma < A$ where $A = \sum_{n=1}^{\infty} [n^{-1} - (n + \frac{1}{2})^{-1}] = 2(1 - \log 2)$, $B = \sum_{n=1}^{\infty} [(n + \frac{1}{2})^{-1} - (n+1)^{-1}] = 2(\log 2 - \frac{1}{2})$.

3.6-4 The integral in (3.2-4) is analytic in z and may be differentiated under the integral sign. By (3.6-10) we obtain the first integral for γ , from which the second follows by choosing e^{-t} as a new variable. Integrate the third integral by parts and use the first.

3.6-5 Take the logarithmic derivative of (3.2-9).

3.7-1 In (3.7-1) replace a, b, n by ka, kb, kn , use Exercise 2.2-8, and proceed as in the proof of Theorem 3.7-1.

3.7-2 By (3.4-4) and Exercise 2.2-8, $\Gamma(1/k) \cdots \Gamma(1 - 1/k) = \lim(1, n)^k k^{kn} / (1, kn) n^{(k-1)/2}$. Apply (2.7-1) to $(1, n) = n!$ and $(1, kn) = (kn)!$.

3.7-3 In Exercise 3.7-1 set $b = z$ and $a = 1/k$ and use Exercise 3.7-2.

3.7-4 Use Exercise 3.5-5 and (3.7-2).

3.7-5 Take the logarithmic derivative of (3.7-2), set $z = \frac{1}{2}$, and use (3.6-10) to obtain $\psi(\frac{1}{2})$. Then apply Exercise 3.6-5 to get $\psi(\frac{3}{2})$.

3.8-1 Use Theorem 3.8-2 and note that $|(x + iy)^{x+iy-1/2}| \sim |(iy)^{x+iy-1/2}| e^x \sim |y|^{x-1/2} e^{x-\pi|y|/2}$.

3.9-1 (a) Set $z = \frac{1}{2}$ in (3.9-3).

(b) Set $z = \frac{1}{6}$ in (3.9-3) and use (3.2-8).

(c) Replace z by $\frac{1}{2} - z$ in (3.9-3).

(d) Set $z = iy$ in (3.9-3), use (3.2-8), and note that $\Gamma(-iy) = [\Gamma(iy)]^*$ by (3.2-7).

(e) Set $z = iy$ in (c) and note that $\Gamma(\frac{1}{2} - iy) = [\Gamma(\frac{1}{2} + iy)]^*$ by (3.2-4).

3.9-2 Use (3.9-3) to replace all sines by gamma functions, and apply Exercise 3.7-3 to $\Gamma(nz/\pi)$ and $\Gamma[n(n^{-1} - z/\pi)]$.

3.9-3 Use (3.7-2) to express $\Gamma(\frac{1}{3})$ in terms of $\Gamma(\frac{1}{6})$ and $\Gamma(\frac{2}{3})$, and (3.9-3) to express $\Gamma(\frac{2}{3})$ in terms of $\Gamma(\frac{1}{3})$.

3.9-4 The fraction is $(\frac{1}{6}, n)/(\frac{1}{3}, n)$. Use (3.4-5) and Exercise 3.9-3.

3.9-5 Divide by $\Gamma(z)\Gamma(z + \frac{1}{2})$. Use (3.7-2) on the left and (3.9-3) on the right.

3.9-6 See Copson (1935, Section 9.6).

3.9-7 Write $n^4 + a^4 = n^4(1 + ia^2/n^2)(1 - ia^2/n^2)$ and use (3.9-5).

3.10-1 Since $\log \Gamma$ is convex on $\mathbb{R}_+$, its derivative ψ is increasing. By Exercises 3.7-5 and 3.6-3, $\psi(\frac{1}{2}) > 0$.

3.11-1 Use (2.3-8) and (3.11-4).

3.11-2 Substitute (3.11-1), choose $s = t(1-x)$ as a positive variable of integration, and use (3.2-4).

Chapter 4

4.2-1 See Apostol (1974, Section 11.21).

4.2-2 (i) Set $a = b = \frac{1}{2}$.

(ii) Set $\theta = \frac{1}{2}\varphi$. Equation (3.7-2) follows by (4.2-1) for $z \in \mathbb{C}_+$ and by analytic continuation for $2z \in U$.

4.2-3 In (4.2-1) use the log-convexity of Γ .

4.2-4 Set $u = \tanh^2 x$ and apply (4.2-2).

4.2-5 Set $t^n = u$ to show $I_n = n^{-1}B(1/n, \frac{1}{2}) = \pi^{-1/2}n^{-1} \cos(\pi/n)\Gamma(1/n)\Gamma(\frac{1}{2} - 1/n)$, where the second equality follows from (3.9-3). Apply Exercise 3.9-3.

4.2-6 Set $t = e^x$ and use (4.2-4) and (3.9-3).

4.2-7 The area is $4 \int_0^1 (1-x^a)^{1/a} dx = 4a^{-1}B(a^{-1}, a^{-1} + 1) = 2a^{-1}B(a^{-1}, a^{-1})$.

4.2-8 The element of arc length is $ds = (2a^2 \sec 2\theta)^{1/2} d\theta$, and the perimeter is four times the integral of this expression from $\theta = 0$ to $\theta = \pi/4$. Use (4.2-3) and (3.9-3) with $z = \frac{1}{4}$.

4.2-9 Set $t = ux + (1-u)y$ and use (4.2-2).

4.2-10 Choose $v = u(1+x)/(u+x)$ as a new integration variable and use Cauchy's theorem to deform the path of integration in the v plane.

4.2-11 In the term proportional to u^{v-1} choose $1/u$ as a new integration variable. Recombine the two terms and apply (4.2-4).

4.2-12 Rewrite the integral as half the integral over the real line, set $e^{2t} = s$, and use (4.2-4).

4.2-13 By Theorem 3.3-1 the residue is $(-1)^n \Gamma(y)/\Gamma(y-n)n!$. Use (3.2-8).

4.2-14 Use (3.8-10).

4.3-1 The absolute value of the integral in (4.3-4) cannot exceed the integral of the absolute value of the integrand. By (4.3-1), $B(b_1, \dots, b_k) = 1/(k-1)!$ if $b_1 = \dots = b_k = 1$.

4.3-2 Use (3.5-1) with $f = \log \Gamma$, $x_i = b_i$, and $u_i = 1/k$.

4.3-3 Since the integral is $B(b_1 + m_1, \dots, b_k + m_k)$, the right side can be reduced to $B(b_1, \dots, b_k)$ by using (4.3-1) and (3.2-9).

4.3-4 Use Exercises 3.7-2 and 3.7-3.

4.3-5 The Jacobian is $\partial(u_1, u_2)/\partial(\theta, \varphi) = 4 \sin^3 \theta \cos \theta \sin \varphi \cos \varphi$. The integrals over θ and φ are multiples of $B(b_1 + b_2, b_3)$ and $B(b_1, b_2)$.

4.3-6 Set $x_i^2 = a_i^2 u_i$. The volume integral over positive values of $x_1, \dots, x_n$ then becomes an integral over the simplex E_n . By (4.3-4) its value is $2^{-n} a_1 \cdots a_n B(\frac{1}{2}, \dots, \frac{1}{2}, 1)$, where $\frac{1}{2}$ occurs n times.

4.3-7 Find the volume V from Exercise 4.3-6. The surface area is dV/dr .

4.3-8 In cylindrical coordinates the axially symmetric volume element in $\mathbb{R}^n$ is dx_n times the spherically symmetric volume element in $\mathbb{R}^{n-1}$. Let $x_n = r \cos \theta$ and $\rho = r \sin \theta$, note that $\partial(x_n, \rho)/\partial(r, \theta) = r$, and use Exercise 4.3-7 with n and r replaced by $n-1$ and ρ .

4.3-9 Proceed as in Exercise 4.3-6. The moment of inertia is $\rho abc(a^2 + b^2) B(\frac{3}{2}, \frac{1}{2}, \frac{1}{2}, 1)$, and $\rho = 3M/4\pi abc$ by Exercise 4.3-6.

- 4.4-1 Use (2.3-9) and the definition of u_k .
 4.4-2 Expand each integrand in powers of ux and integrate term by term using (4.4-8).
 4.4-3 Proceed as in Exercise 4.4-2.
 4.4-4 Proceed as in Exercise 4.4-2 and use (2.2-7).
 4.4-5 Set $(x_i/a_i)^{b_i} = Tv_i$ and use (4.14) with $\varphi(u) = f(Tu)$. Set $Tu = t$.
 4.4-6 Use Exercise 4.4-5.
 4.4-7 Use Exercises 4.4-5 and 3.2-4.
 4.4-8 Use (4.4-4) and (3.10-7).

Chapter 5

- 5.1-1 Set $t = ux + (1 - u)y$ and apply (5.1-1).
 5.1-2 Set $u = \cos^2 \theta = (1 + t)/2$ in (5.1-1).
 5.1-3 Set $\beta' = \beta$ and $\theta = \frac{1}{2}\varphi$ in Exercise 5.1-2 and apply Exercise 4.2-2.
 5.1-4 In Exercise 5.1-1 set $\beta' = \beta$ and replace (x, y) by $(X, Y) = (x + ye^{i\varphi}, x + ye^{-i\varphi})$. Use Cauchy's theorem to deform the straight path of integration into the arc of a circle with center x . Set $t = x + ye^{i\varphi}$ and hence $(X - Y)^{-2}(t - Y)(X - t) = (2 \sin^2 \theta)^{-1}e^{i\varphi}(\cos \varphi - \cos \theta)$. Apply Exercise 4.2-2.
 5.2-1 For the right-hand inequality use Exercise 3.5-7 in the integrand of (5.2-1) and integrate by means of (4.4-8). The left-hand inequality is $f(\int u \cdot x \, d\mu) \leq \int f(u \cdot x) \, d\mu$; see Rudin (1974, Section 3.3). If f is concave, $-f$ is convex.
 5.2-2 Note that $|F(b, z)| \leq \sup |f(\zeta)| |\mu_b|(E)$ and use (4.4-4).
 5.3-1 Use the chain rule and (5.3-3).
 5.4-1 The first equality is (5.4-1). The second is derived by differentiating (5.1-1) under the integral sign.
 5.4-2 Write $(x - y)^2 F''$ as an integral over $[0, 1]$ and integrate by parts twice.
 5.4-3 Differentiation of the product allows the general case to be deduced from the case $\alpha = 0$, which follows from Exercises 5.4-1 and 5.4-2.
 5.4-4 Substitute $v = (x - y)^2 \varphi$ in (5.4-1).
 5.4-5 In Exercise 5.4-4 set $\alpha = 1 - \beta - \beta'$ and replace β by $1 - \beta'$ and β' by $1 - \beta$.
 5.4-6 Let $z = (x + t, x - t)$. Then $(x, t) \in \sigma$ is equivalent to $z \in I^2$, and the differential equation for v is equivalent to (5.4-2) with $k = 2$. Equation (5.2-2) implies $v(x, 0) = f(x)$. Since the second derivatives of v are finite on σ by Theorem 5.3-2, the differential equation for v requires $(\beta + \beta')v_t(x, 0) = (\beta - \beta')v_x(x, 0)$. Thus $v_t(x, 0) = 0$ if $\beta = \beta'$.
 5.4-7 Direct substitution shows that v_0 is a solution, and hence so is v if γ is independent of z . For the case in which the endpoints depend on z , see Carlson (1963, p. 466).
 5.4-8 Write each term on the left as an integral over $[0, 1]$ and integrate by parts.
 5.5-1 Prove by induction using (5.5-4).
 5.5-2 In (5.5-4) set $z_j = x + jh$, $j = 1, \dots, n$, and $y = x + (n + 1)h$. Use induction.
 5.5-3 In (5.5-4) replace x by $x + \frac{1}{2}(n + 1)h$, y by $x - \frac{1}{2}(n + 1)h$, and set $z_j = x + \frac{1}{2}(n + 1)h - jh$, $j = 1, \dots, n$. Use induction.
 5.5-4 Write $I^n(I^m f)$ as a repeated integral, change the order of integration by Fubini's theorem, and reduce the inner integral to (4.2-2) by change of variable.
 5.5-5 Use Exercise 5.1-1 and change the order of integration by Fubini's theorem.
 5.5-6 Use Exercise 5.1-1. When $J^{v+1}f$ is differentiated, the term which comes from the upper limit of integration vanishes because $\operatorname{Re} v > 0$.

5.6-1 Use (5.6-8) in the integrand of $\int (u \cdot z) f(u \cdot z) d\mu_b(u) = H(b, z)$.

5.6-2 Make a multinomial expansion of $(\sum_{i=1}^k u_i)^n = 1$ and use (5.6-7).

5.7-1 In (5.7-6) set $f = g$ and $\lambda = 0$.

5.8-1 Define $s(\lambda) = S(b, z + \lambda)$, whence $s'(\lambda) = \sum D_i S(b, z + \lambda)$. Thus the second equation of (5.8-5) is equivalent to $s'(\lambda) = s(\lambda)$, while (5.8-3) is equivalent to $s(\lambda) = e^{\lambda s(0)}$. Each property implies the other.

5.8-2 In Exercise 5.4-2 set $\beta = v - \frac{1}{2}$, $x = -y = iz$, and $F = S$. Use (5.8-23).

5.8-3 In (5.5-8) set $f(x) = e^x$ and $a = 0$. Use (5.8-2).

5.8-4 In Exercise 5.2-1 set $f(y) = e^y$.

5.8-5 Note that $f(u \cdot x) \leq \exp[u \cdot \lambda(x)]$ if f is log-convex, with reversed equality if $1/f$ is log-convex. Integrate with respect to Dirichlet measure. The remaining inequalities follow from Exercise 5.8-4.

5.8-6 Set $F = S$ in (5.5-3).

5.8-7 Use (5.8-23) and Exercise 5.1-3 with $F = S$. Note that $\sin(x \cos \varphi)$ is odd about $\varphi = \frac{1}{2}\pi$.

5.8-8 In Exercise 5.4-3 set $x = \delta + \sigma z$, $y = \delta - \sigma z$. To obtain the second equation choose $\delta = 0$, $\sigma = i$, $\beta = v - \frac{1}{2}$, $\alpha = v + \lambda$, $F = S$, and apply (5.8-23).

5.8-9 Use Exercises 5.8-2 and 5.8-8.

5.8-10 By induction the n th forward difference (see Exercise 5.5-2) of $f(x) = e^x$ is $\Delta^n f(x) = e^x(e^h - 1)^n$. Use Exercise 5.5-2 and replace x, h by $nx, y - x$. To obtain the second equation, replace x, y in the first by $ix, -ix$.

5.8-11 By substituting $\varphi = x^\rho S$ and differentiating, it is sufficient to prove the case $\rho = 0$. Let $y = \sigma x$ and $z = \tau x$, whence $d/dx = \sigma \partial/\partial y + \tau \partial/\partial z$. From this and $(\partial/\partial y + \partial/\partial z)S = S$, find $\partial S/\partial y$ and $\partial S/\partial z$ in terms of S and dS/dx . Substitute in the Euler-Poisson equation for S .

5.8-12 Choose the polar axis in the direction of k , and let $k \cdot x = Kr \cos \theta$. Use Exercise 4.3-8, Fubini's theorem (Appendix B.2), and Exercise 5.1-3 with $f(x) = e^x$.

5.9-1 In (5.5-8) set $f(x) = (1 - x)^{-\alpha}$ and $a = 0$. Use (5.9-2) and (5.2-4).

5.9-2 In (5.9-21) set $y = 1$, $\alpha' = \gamma - \alpha$, $\beta' = \gamma - \beta$, replace x by $1 - x$, and use (5.9-11). See the Remark following Proposition 5.9-8 concerning restrictions on the parameters.

5.9-3 In Exercise 5.9-2 multiply both sides by $(1 - x)^{\alpha + \beta - \gamma}$ and compare coefficients of x^n to get the first form of Saalschütz' theorem. Use (2.2-8) to obtain the second form.

5.9-4 By (5.8-7) and (5.8-3) the left side is $S(\beta, \gamma - \beta; y - xy, -xy)$. Use (5.8-4) and (5.9-11).

5.9-5 Multiply (5.9-7) by w_i and sum i from 1 to k . Use (5.9-5), replace t by $t + 1$, and divide by $t + 1$.

5.9-6 Write (5.9-8) once with index i and once with index j , and eliminate R_i or R_{i-1} to prove the first or second relation, respectively.

5.9-7 Multiply (5.9-7) by $w_i z_i^{-1}$, sum i from 1 to k , and use (5.9-5) to prove the first relation. For the second replace t by $t + 1$ in (5.9-7), multiply by $w_i z_i$, sum i from 1 to k , and use (5.9-6).

5.9-8 Solve (5.9-5), (5.9-6), and the second equation of Exercise 5.9-7 (with $k = 3$) for $R_i(b + e_i, z)$. Replace t by $t - 1$.

5.9-9 Substitute the result of Exercise 5.9-8 in (5.9-7).

5.9-10 Iterate (5.9-7) and use (5.9-25).

5.9-11 In Exercise 5.2-1 set $f(y) = y^t$.

5.9-12 The first property follows from $u \cdot (\lambda z) = \lambda(u \cdot z)$, and the second from (5.7-4) with $f(x) = -\log(1 - x)$ and $\lambda = 0$. By Theorem 5.9-2 we may differentiate (5.9-1) with respect to t under the integral sign. The differential properties are special cases of (5.3-3) and (5.3-4). The case $b_1 = b_2 = 1$ follows from (5.5-2), and the inequalities from Exercise 5.2-1.

- 5.9-13** In (5.5-3) choose $f(x)$ to be x' , $t \neq 0$, or $\log x$.
- 5.9-14** Use (5.8-4), the homogeneity of R_n , and (5.9-25).
- 5.9-15** By (5.7-1) and (5.9-21) the integral is $B(x, y)R_{-x-y}(x, y; 1 + z, z) = B(x, y)(1 + z)^{-x}z^{-y}$.
- 5.9-16** Use (5.9-8), (5.9-24), and (5.9-25).
- 5.9-17** In Exercise 5.4-2 set $F = R_t$ and replace β by $\beta - 1$.
- 5.9-18** Use Exercise 5.3-1.
- 5.9-19** Set $\sin^2 \theta = u$ and apply (5.9-1).
- 5.9-20** Set $(l^2, m^2, n^2) = (u_1, u_2, u_3)$. The Jacobian of (u_1, u_2) with respect to (θ, φ) is $4 \sin \theta (u_1 u_2 u_3)^{1/2}$. Use (5.9-1).
- 5.9-21** Set $x = \lambda^{-1/2} r \sin \theta \cos \varphi$, $y = \mu^{-1/2} r \sin \theta \sin \varphi$, $z = \nu^{-1/2} r \cos \theta$. Use Exercises 5.9-20 and 3.2-4.
- 5.9-22** See Carlson (1972c).
- 5.9-23** The sum is ${}_3F_2(-n, n, 1; \frac{1}{2}, \frac{3}{2}; 1) = 1/(1 - 4n^2)$ by Exercise 5.9-3.
- 5.9-24** Let $f(x) = x^n \log x$, whence $f^{(n+1)}(x) = n!/x$ and $f^{(k-1)}(x) = n!(-1)^{k-n}(k - n - 2)! \cdot x^{n-k+1}$. Use Exercise 5.5-1.
- 5.9-25** Set $\theta + \varphi = \psi$ and show by periodicity and evenness that the integral over the interval $\varphi \leq \psi \leq \varphi + 2\pi$ is twice the integral over $0 \leq \psi \leq \pi$. Use Exercise 5.1-3.
- 5.10-1** To justify taking the limit of the series term by term, choose a positive constant $r > |x|$ and use Exercise 2.2-10 and the Weierstrass M test to prove uniform convergence for $|\alpha| \geq r$.
- 5.10-2** Since $\xi(t) \rightarrow x$ as $t \rightarrow \infty$ we can choose r large enough so that ξ maps the annulus $A = \{t \in \mathbb{C} : |t| \geq r\}$ into the disk $D = \{\xi \in \mathbb{C} : |\xi| \leq \frac{1}{2}r\}$. Then (5.10-3) with x replaced by $\xi(t)$ is still majorized on A by (5.10-5), and (5.10-8) is proved. Next, choose r large enough so that ξ maps A into D^k . Then (5.10-7) holds with z replaced by $\xi(t)$.
- 5.10-3** Use (5.8-23), (5.8-3), and (5.10-11).
- 5.10-4** Use Exercise 5.10-3, (5.9-22), and (3.7-2).
- 5.10-5** Use Exercise 5.8-10, (5.8-3), and (5.10-11).
- 5.11-1** In (5.11-5) set $n = 2v$, $b = (v + \frac{1}{2}, v + \frac{1}{2})$, $z = (ix, -ix)$. Use (5.8-23) and (5.9-22).
- 5.12-1** Set $t = xu^{1/2}$ in the first integral and $t = (x^2 + v)^{1/2}$ in the second. Use (5.8-1) and (5.12-7).
- 5.12-2** Set $t = xu$ in the first integral and $t = x + v$ in the second. Use (5.8-1) and (5.12-7).
- 5.12-3** Set $t = x - s$ and use (5.12-7).
- 5.12-4** Set $t = xe^{-s}$ and use (5.12-7).

Chapter 6

- 6.2-1** Set $\beta' = y = 1$ in (6.2-3).
- 6.2-2** Substitute (6.2-1) for R_m . The resulting multiple sum is again of the type that occurs in (6.2-1).
- 6.2-3** Use (6.2-16) with (5.9-7) for E_n or (5.9-8) for C_n .
- 6.2-4** Use (6.2-3) or Exercise 5.9-13.
- 6.2-5** Set $\alpha = -n$ in (5.9-11) and multiply by (γ, n) . Since both sides are polynomials in β and γ , the resulting equation holds for all complex β, γ . Divide by $(\gamma, n) \neq 0$.
- 6.2-6** If $w = b/c$ is held fixed as $\|b\| \rightarrow 0$, $R_n(b, z) \sim \sum w_i z_i^n$ by (6.2-17) and $1/\Gamma(c) \sim c$ by (3.3-3). Since $R_n(b, z)/\Gamma(c)$ is entire in b and z by Theorem 6.2-2, its value is determined uniquely by b and z .
- 6.2-7** Since the case $\|b\| = 0$ is trivial by Exercise 6.2-6, assume $\|b\| > 0$. Divide (6.2-24) by $|\Gamma(c + n)|$ and use Exercise 3.4-5.

6.2-8 Since the case $n = 0$ is trivial, assume $n > 0$ and set $b = cw$ in (6.2-24). Note that $(\|cw\|, n) \leq 2|c|(\frac{1}{2}\|w\|, n)$ and $|(c, n)| \geq |c|(1 - |c|) \cdots (n - 1 - |c|) \geq |c|(\frac{1}{2}, n - 1)$.

6.2-9 Use (6.2-7) and (2.3-8).

6.2-10 Since the case $n = 0$ is trivial, assume $n \geq 1$, which implies $s \geq 1$. Refer to the proof of Theorem 6.2-6 and write $|(\beta, m_i)| = |(\beta, s)(\beta + s, m_i - s)| \leq |(\beta, s)|(|\beta| + s, m_i - s)$. Since $(s - 1)! \leq (|\beta| + 1, s - 1)$ and $s \leq |\beta| + m_i$, we have $|(\beta, m_i)|s! \leq |(\beta, s)|(|\beta| + 1, s - 1)(|\beta| + s, m_i - s) = |(\beta, s)|(|\beta| + 1, m_i)$. For every $j \neq i$, $|(\beta, m_j)| \leq (|\beta| + 1, m_j) \leq (|\beta| + 1, m_j)$. By (6.2-7), $|(k\beta, n)R_n(b, z)| \leq (n!/s!)|(\beta, s)z^n| \sum (|\beta| + 1, m_1) \cdots (|\beta| + 1, m_k)/m_1! \cdots m_k!$. Use (2.3-5) to obtain the first inequality and multiply by $|\Gamma(\beta)/\Gamma(k\beta + n)|$ to get the second.

6.2-11 Use Exercises 6.2-10 and 3.4-5.

6.2-12 Use Example 6.2-10 and note that $s! = \Gamma(s + 1) > \Gamma[1 + (n - 1)/k]$ by (3.10-2) since $s > (n - 1)/k \geq 1$. By (3.10-7), $|\Gamma(k\beta + n)| > \Gamma(\operatorname{Re} k\beta + n) \exp(-\pi |\operatorname{Im} k\beta|/2) > \Gamma(n - kr) \exp(-\pi kr/2)$ since $\operatorname{Re} k\beta + n \geq n - kr \geq k + 1 \geq 2$. Also, $s - 1 \leq (n - 1)/k < s$ leads to $1 \leq -r + (n - 1)/k < \operatorname{Re} \beta + s \leq r + 1 + (n - 1)/k$ and hence, by (3.10-5), to $|\Gamma(\beta + s)| \leq \Gamma(\operatorname{Re} \beta + s) \leq \Gamma[r + 1 + (n - 1)/k]$. For this last inequality note that $1 \leq x \leq y$ and $y \geq 2$ imply $\Gamma(x) \leq \Gamma(y)$.

6.2-13 Use (6.2-7). Expand $(z_i - z_j)^2$ to reduce the second equation to the first.

6.2-14 Expand z^n to second order in $1 - z/z$. Use (5.9-4), (6.2-2), and Exercise 6.2-13.

6.3-1 All three factors are differentiable on the cut plane by Corollary 6.3-4. When s describes a closed path encircling the cut, the product returns to its initial value.

6.3-2 Use Exercise 6.2-9 and (5.9-20). The latter equation, divided by $\Gamma(\beta + \beta')$, holds more generally by Corollary 6.3-4 and the permanence of functional relations.

6.3-3 Use (6.3-4) to represent the right side of (5.5-14). Set $v = -n$, use Exercise 6.3-2 and Corollary 6.3-5, and evaluate the integral by Cauchy's integral formula.

6.3-4 Set $b = cw$ in (6.3-1). Use Exercise 6.2-8 and the Weierstrass M test to prove uniform convergence in c on $0 < |c| \leq \frac{1}{2}$. Take the limit of the series term by term using (6.2-17).

6.3-5 The double series converges absolutely by Exercise 2.4-10. Change the order of summation and use (6.2-7), (6.2-6), and (6.3-6). Set $(\beta, \beta', \gamma - \beta - \beta') = (b_1, b_2, b_3)$, $x = 1 - z_1/z_3$, $y = 1 - z_2/z_3$, and use the symmetry of $R_{-s}(b_1, b_2, b_3; z_1, z_2, z_3)$ in 1 and 3 to obtain the transformation given by Erdélyi (1953, 5.11(2)).

6.3-6 Proceed as in the proof of Corollary 6.3-5, using Corollary 6.3-7.

6.4-1 In (6.4-2) set $b = (\beta, \gamma - \beta)$, $z = (x - xy, x)$, $\lambda = 1 - x$. Use Exercise 6.2-5.

6.4-2 See Carlson (1970b, Theorem 1).

6.5-1 The proof is similar to that of Transformation 6.5-1. Apply the binomial theorem to make the first argument zero and remove this argument by (6.2-6). Rearrange the Appell symbols and perform the summation by Exercise 6.2-2.

6.5-2 Apply Exercise 6.2-5 to the three forms of R_n given in Transformation 6.5-1 to obtain the first, third, and sixth ${}_2F_1$ polynomials. The others follow from these by the symmetry of R_n .

6.5-3 In Exercise 6.5-2 replace (x, y) by $(1 - x, 1)$.

6.6-1 Divide (6.6-1) by t^{n+1} and integrate term by term.

6.6-2 Use (6.6-8) and (6.6-17).

6.6-3 In Exercise 5.8-10 set $y = 0$. Use (5.8-4) and (6.6-17).

6.6-4 Write $at^2 + 2bt + c$ in the form $c(1 - tr)(1 - ts)$ and use (6.6-1).

6.6-5 If R_{2m} is the input resistance, then $R_2 = 2$ and $R_{2m+2} = 1 + (1 + R_{2m}^{-1})^{-1}$. By induction $R_{2m} = F_{2m+1}/F_{2m}$, where F_n is a Fibonacci number.

- 6.6-6** Multiply (6.6-1) by the same generating function with b, z replaced by β, ζ . Equate coefficients of t^n .
- 6.6-7** Proceed as in Exercise 6.6-6 with $\zeta = z$. Alternatively set $k = \kappa$ and $z = \zeta$ in Exercise 6.6-6 and apply Theorem 5.2-4.
- 6.6-8** Multiply (6.6-1) by the same generating function with z replaced by $-z$. Equate coefficients of t^{2n} .
- 6.6-9** Set $c = n + 1$ in (6.3-4). Then (6.6-5) can be used for every $s \in \gamma$ by Corollary 6.3-5 and the permanence of functional relations.
- 6.6-10** Use (6.6-1) and Exercise 6.2-5.
- 6.6-11** Multiply corresponding sides of (6.6-8) and (6.6-9) and equate coefficients of t^n .
- 6.6-12** In Exercise 6.6-7 choose $\beta_j = \delta_{ij}$.
- 6.6-13** Use Exercise 6.6-12 and (5.9-6).
- 6.6-14** In Exercise 6.6-13 choose $b = (-1, \dots, -1)$ or $b = (1, \dots, 1)$.
- 6.6-15** Set $f(s) = \log s$ in Exercise 6.6-9.
- 6.6-16** Use (6.6-5).
- 6.6-17** Set $t = s + 2m\pi$. Use Exercise 5.10-5 and (6.6-5).
- 6.7-1** Use (6.7-5) and (6.7-21).
- 6.7-2** Use (6.2-24) with (6.7-5) and (6.7-21).
- 6.7-3** Use (6.7-21) and Exercise 5.9-13.
- 6.7-4** Factor the quadratic and use the Taylor series of $\log(1 - tx)$.
- 6.7-5** Multiply (6.7-22) by $C_n^v(y)/C_n^v(1)$, antisymmetrize in x and y , sum with respect to n from 1 to N , and use (6.7-20).
- 6.7-6** In Exercise 6.6-9 set $b = (v, v)$, $n = 2v - 1$, $z = (e^{i\theta}, e^{-i\theta})$, and subsequently $f(s) = s^{n+2v-1}$. The requirement that $\text{con}(z) \subset I(\gamma)$ can be removed by Cauchy's theorem.
- 6.7-7** Use Exercise 6.6-7, with $b = \beta = (\frac{1}{2}, \frac{1}{2})$, and Exercise 6.7-3.
- 6.7-8** Use Exercise 6.6-8 with $b = (v, v)$ and $z = (e^{i\theta}, e^{-i\theta})$.
- 6.7-9** In (6.4-2) set $b = (v, v)$, $z = (\cot \varphi + i, \cot \varphi - i)$, $\lambda = \cot \theta - \cot \varphi$. Use (6.7-21) and the homogeneity of R_n .
- 6.7-10** Set $v = \frac{1}{2}$ in Exercise 6.7-9. The first equation is the case $\varphi = 2\theta$ and the second is the case $\theta = \varphi + \frac{1}{2}\pi$.
- 6.7-11** In Exercise 6.6-13 set $b = (v, v)$ and $z = (e^{i\theta}, e^{-i\theta})$.
- 6.7-12** Use (6.7-2) and (6.7-19).
- 6.7-13** In (5.10-9) set $b = (v, v)$ and $1 + \zeta(t)/t = (e^{ix/t}, e^{-ix/t})$.
- 6.8-1** Use (6.8-17), (6.8-15), and (3.9-3). Both sides of the second equation are entire in t and v by Theorem 6.8-2.
- 6.8-2** Use (6.8-17) and Exercise 5.1-3 for the first integral. Note that $\text{Re}(\cos \theta)$ has the same sign as $\text{Re}(e^{\pm i\theta})$ even if θ is complex. Hence $x \in \mathbb{C}_>$ implies $x \pm (x^2 - 1)^{1/2} \in \mathbb{C}_>$ for both choices of sign. For the second integral use Exercise 6.8-1 and (3.9-3).
- 6.8-3** In Exercise 5.1-4 set $\beta = v$, $x = 0$, $y = 1$, and $f(x) = x^t$. Even if $t \notin \mathbb{N}$, f is holomorphic on the right half-plane, and the deformation of contour used in solving Exercise 5.1-4 is still permissible provided $0 < \theta < \frac{1}{2}\pi$.
- 6.8-4** In (6.3-6) set $a = -\alpha$, $b = (v, v)$, and $z = (1 - te^{i\theta}, 1 - te^{-i\theta})$. Use (6.7-26) on the right side and (6.8-18) on the left.
- 6.8-5** Use Exercise 5.9-13 and the permanence of functional relations. In the first equation set $\theta = \frac{1}{2}\pi$ and replace t by $1 + t$ to obtain the second.
- 6.8-6** Use Exercise 5.9-13 and the permanence of functional relations.
- 6.8-7** Use (6.8-15) and (6.2-2).
- 6.8-8** Replace t by $1/t$ in (6.8-6).

- 6.9-1** Set $x = i$ in Theorem 6.9-1 and use (6.7-21) and (6.7-5).
- 6.9-2** Combine Theorem 6.9-1 with Transformation 6.5-1 to obtain (1) and (2); combine Theorem 6.9-1 with Exercise 6.5-2 to obtain (3). Use Exercise 6.2-5 and (2) to get (4) and (5).
- 6.9-3** By (6.6-1) the generating function is $[1 - t(x + y) + t^2xy]^{-\beta}$. Expand first in powers of $t(x + y) - t^2xy$ and then, using the binomial theorem, in powers of t . Change the order of summation to find the coefficient of t^m .
- 6.9-4** In Exercise 6.9-3 set $x = e^{i\theta}$, $y = e^{-i\theta}$, and $\cos \theta = z$.
- 6.9-5** Proceed as in the proof of Transformation 6.9-3, starting with $R_t(\beta, \beta, \delta; 1 + w, 1 - w, 1)$ and using (6.2-6).
- 6.9-6** Use (6.4-2), with $\lambda = x$, and Theorem 6.9-1.
- 6.9-7** Use Exercise 6.9-6 in the integrand, integrate by (4.4-8), and use Exercise 6.9-6 again.
- 6.9-8** Apply (6.5-1) to (6.9-8) and (6.9-9).
- 6.9-9** Use (6.7-21) and Exercises 6.9-8 and 6.5-2.
- 6.9-10** Let $\beta \rightarrow 0$ in (6.9-8) and (6.9-9). Use (6.7-28) and Exercise 6.5-2.
- 6.9-11** Set $\beta = 1$ in (6.9-8) and (6.9-9). Use Exercises 6.7-3 and 6.5-2.
- 6.9-12** Use (6.6-23) and Exercise 6.9-3.
- 6.9-13** Equate coefficients of t^n in $(1 - t\omega)^{-\beta} \cdots (1 - t\omega^k)^{-\beta} = (1 - t^k)^{-\beta}$.
- 6.9-14** The vertices are $(z_1, \dots, z_k) = (\lambda + x\omega, \dots, \lambda + x\omega^k)$, where $\omega = e^{i2\pi/k}$. Use (6.3-1) and Exercise 6.9-13. For the two special cases choose $f(x)$ to be e^x and x^{-a} , and use Exercise 2.2-8 and (2.4-9).
- 6.9-15** Use (5.8-3), (5.8-4), (6.2-6), and Theorem 6.9-1.
- 6.9-16** Use (6.9-15).
- 6.9-17** Use Exercise 6.8-5 and (6.9-7) to get the first equation. For the others set $x = \sin \theta$ in (6.9-15) and then $\theta = \pi/2, \pi/4, \pi/6$ and $\theta = i\varphi$.
- 6.9-18** The solution of Exercise 6.2-14 remains valid if $z^s \in \mathbb{C}_0^k$.
- 6.9-19** Given $r > 0$ show that the series converges uniformly for $|c| \leq r$ and $|x| \leq r$ by using (6.2-30) and the Weierstrass M test.
- 6.9-20** Differentiate term by term as in Exercise 2.4-7. Use (6.9-18).
- 6.9-21** If the first member of (6.9-24) is written as a power series, the coefficient of x^m is 0 if $m < n$. Choose $m - n$ as a new summation index to prove (6.9-24) if $n \in \mathbb{N}$. The case $-n \in \mathbb{N}$ is reduced to the previous case by setting $n = -n'$. Use (6.9-18) and (6.9-22) to prove (6.9-25).
- 6.10-1** In (6.10-21) set $t = -\alpha$, $x = 1 - 2z$, $y = 1$, and use (5.9-11).
- 6.10-2** In (6.10-6), $x_0 = 1$, $y_0 = \sqrt{2}$, $x_1 = 1.207107$, $y_1 = 1.189207$, $x_2 = 1.198157$, $y_2 = 1.198124$, $x_3 = y_3 = M = 1.198140$.
- 6.10-3** In Exercise 4.2-5 set $n = 4$ and $t = u^{1/2}$. Use Exercise 6.10-2.
- 6.10-4** The differential equation is $\ddot{\theta} + (2\pi/T_s)^2 \sin \theta = 0$ with initial conditions $\dot{\theta} = 0$ and $\theta = \varphi$. A first integral, expressing conservation of energy, is $\frac{1}{2}\dot{\theta}^2 = (2\pi/T_s)^2(\cos \theta - \cos \varphi)$, and the period is $T = 4 \int_0^{\varphi} (\dot{\theta})^{-1} d\theta$. Set $\sin^2 \frac{1}{2}\theta = u \sin^2 \frac{1}{2}\varphi$ and use (5.9-1). By Exercise 6.10-2, $T/T_s = \sqrt{2} R_K(1, 2) = 1.18034 \dots$
- 6.10-5** Express both transformations in terms of R_K by (6.10-10). Use homogeneity, symmetry, and (6.10-5).
- 6.10-6** Use Exercise 6.6-4 and (6.10-11).
- 6.10-7** Use (6.10-11) and (6.7-26).
- 6.10-8** In (6.10-11) set $\beta = \frac{1}{2} + \alpha$ and replace (x, y) by $(x + y, x - y)$.
- 6.10-9** Use (6.6-23) and (6.10-11).
- 6.10-10** Use (6.7-21), (6.10-1), and Exercise 6.2-5.
- 6.10-11** Use Exercise 6.7-3 and (6.10-11).
- 6.10-12** Use (6.10-1), (6.3-3), and (5.9-22).

6.10-13 From $x_0 = 1$ and $y_0 = \sqrt{2}$ compute $\sigma_0 = 1.2761424$, $\sigma_1 = 1.2734109$, $\sigma_2 = 1.2732501$, $\sigma_3 = 1.2732402$. If $A_n = \sigma_n + (\sigma_n - \sigma_{n-1})/15$, then $A_1 = 1.2732288$, $A_2 = 1.2732394$, $A_3 = 1.2732395 = 1/0.7853982$.

6.11-1 Use Exercises 5.9-25, (6.10-8), and (6.11-4).

Chapter 7

7.1-1 Use (6.2-3) and Exercise 2.2-7.

7.1-2 Multiply the ${}_0F_1$ series, collect coefficients of t^n , and use Exercise 7.1-1.

7.1-3 Use (2.2-12), (6.2-3), and Exercise 6.5-2.

7.1-4 Use (5.9-11) and (5.9-21).

7.1-5 Apply (6.4-2) to $p_n(x)$ and (5.9-4) to $y^{n+1}q_n(y)$.

7.1-6 Use Exercise 5.9-16.

7.1-7 Multiply the first equation in Exercise 7.1-6 by $q_n(y)$ and the second by $p_n(x)$, subtract, and sum with respect to n .

7.1-8 Express the derivatives as R functions by Exercise 5.9-18 and use (5.9-25). By (5.9-21), $(x-r)^{-\beta}(x-s)^{-\alpha}q_n$ has the same form as p_n except for replacement of $-n$ by $1 + \alpha + \beta + n$, which does not change the differential equation.

7.1-9 Use Exercise 7.1-2 and (6.10-11). See the Remark following Theorem 6.2-6.

7.1-10 In Exercise 7.1-9 set $\delta = 1$, $x = t \cos \theta$, $y = it \sin \theta$. Use (6.9-18) and (6.9-21).

7.1-11 In Exercise 7.1-6 set $y = x$, multiply the first equation by $q_n(x)$ and the second by $p_n(x)$, and subtract. This shows that the first member of the equation to be proved is a constant independent of n . Set $n = 0$ to find the constant.

7.1-12 Use Exercise 6.9-8.

7.1-13 Terms with a common value of $m + n$ can be summed by Exercise 7.1-1. If $\gamma = \delta$, use (6.10-11).

7.2-1 Set $\alpha = \beta = v - 1$, $r = -1$, $s = 1$ in (7.2-9) and use (6.10-12). See Corollary 6.3-7 for the condition on v .

7.2-2 Use (7.2-10) and (5.2-4).

7.3-1 Set $\theta = \varphi$, $\psi = 0$, $\cos \theta = x$ in (7.3-11).

7.3-2 Combine (6.7-15) and (7.3-11).

7.4-1 Use (6.10-11) and Exercise 5.9-13. For the second equation use (6.10-1) and proceed as in the proof of Lemma 7.4-1.

7.4-2 In (7.4-6) set $\alpha = \beta = v - \frac{1}{2}$. Use (6.10-11), (3.4-5), and (6.7-21). The restrictions on v make all coefficients nonzero.

7.4-3 Copson (1965, Chapter 8) is a helpful guide.

7.4-4 Use (6.10-12), (3.4-5), and Exercise 7.4-3.

7.5-1 For every $\rho > \mu(x)$, $|p_n(x)| \leq \hat{p}_n(\rho)$ and thus $\limsup |p_n(x)|^{1/n} \leq \rho$ by (7.5-7). Hence the upper limit cannot exceed $\mu(x)$.

7.5-2 In (7.2-14) choose γ to be Λ_σ , where $4\rho > 4\sigma > |r - s|$. The integral is majorized by $M\hat{q}_n(\sigma)8\sigma$, since the perimeter of Λ_σ is less than the perimeter (8σ) of the smallest rectangular area which covers Λ_σ . Since σ is arbitrarily close to ρ and $\hat{q}_n$ is continuous, σ can be replaced by ρ .

7.6-1 Let C be a circle with center γ and large radius R . Apply Cauchy's integral formula to the simple closed contour consisting of C described in the positive direction, γ described in the negative direction, and the edges of a cut connecting γ and C . Let $R \rightarrow \infty$.

7.6-2 Apply Theorem 7.6-2 to $f(X)$ with $X = A + Bx$, $r = A - B$, $s = A + B$, $\alpha = \beta = v - \frac{1}{2}$. Use (6.10-12).

7.6-3 Use Exercise 7.6-2 to get the first series, which converges uniformly in $\cos \theta$ on compact subsets of Ω . Replace $\cos \theta$ by $u \cos(\theta + \varphi) + (1 - u) \cos(\theta - \varphi)$, integrate term by term with respect to $\mu_{(v, v)}$, and use (6.11-5).

7.6-4 Set $x = a + bz + cz^{-1}$, $r = a + 2(bc)^{1/2}$, $s = a - 2(bc)^{1/2}$. By (7.5-3), $\mu(x) = \max\{|bz|, |c/z|\} < \rho$. Use Theorem 7.6-2 with $\alpha = \beta = -\frac{1}{2}$ and (7.4-1).

7.7-1 In (7.7-2) set $\alpha = \beta = v - \frac{1}{2}$, $r = -1$, $s = 1$. Replace λ by $i\lambda$ and use (5.8-23) and (6.10-12).

7.7-2 In Exercise 7.7-1 set $x = 0$ and use Exercise 6.9-1.

7.7-3 Multiply the second equation of (7.7-9) by $e^{-in\theta}$ and integrate term by term (see Theorem 7.6-2) over $[0, 2\pi]$. Since the integrand has period 2π , the interval can be changed to $[-\pi, \pi]$. Only the even part of the integrand then contributes to the value of the integral.

7.7-4 Put $\theta = \frac{1}{2}\varphi$ and apply (7.7-6).

7.7-5 In (7.7-12) replace x, r, s by $x + y, (\sqrt{x} + \sqrt{y})^2, (\sqrt{x} - \sqrt{y})^2$ and apply Theorem 6.9-1.

7.7-6 In (7.7-12) set $c = 1$ and replace $-4x, -4r, -4s$ by $A^2 + B^2 - 2AB \cos \theta, (A + B)^2, (A - B)^2$. Use (6.9-18) and (6.10-14).

7.7-7 In (7.7-13) set $A = B$ and $x = \cos \theta$.

7.7-8 Use (7.7-6). Expand by (6.11-4).

7.7-9 Set $f(x) = (1 - 2tx + t^2)^{-\lambda}$ in Exercise 7.6-3. Use Exercise 6.10-12.

7.7-10 Set $f(x) = e^{\lambda x}$ in Exercise 7.6-3. Use (5.8-2) and (5.8-3).

7.7-11 In Exercise 7.7-10 set $\lambda = irs$ and $v = \frac{1}{2} + m$. Use (5.8-23), (5.8-24), and (6.10-18).

7.7-12 Use Exercise 7.6-4 with $a = A$, $b = \frac{1}{2}(B - iC)$, $c = \frac{1}{2}(B + iC)$, $z = e^{i\theta}$.

7.7-13 Use Exercises 7.7-12 and 6.10-12.

7.7-14 Use Theorems 7.6-2 and 5.2-6.

7.7-15 In Exercise 7.7-14 set $A = r^2 + r'^2$, $B = 2rr'$, $x = \cos \theta$. Use (6.10-12). To recover Exercise 6.7-12 use Exercise 6.10-12.

7.8-1 Use (7.8-1) with (6.10-12) and (6.10-13).

7.8-2 Use (7.8-4) with (6.10-18).

7.8-3 Use (7.8-4) with (6.10-12).

7.8-4 Set $n = 0$ in Exercise 7.8-3.

7.8-5 Expand the differential operator by the binomial theorem and use (6.2-3). To obtain (7.8-1) set $x = X - r$ and $y = X - s$ and use the chain rule for differentiation with respect to X .

7.8-6 By (5.9-2), $D^n R_{-c}(b, z) = (-1)^n (c, n) R_{-c-n}(b, z)$. Apply (6.8-15) to both R functions. For Exercise 7.8-5 set $z = (x, y)$, $b = (\beta', \beta)$, and use the homogeneity and symmetry of R_n .

7.8-7 Set $\cos \theta = y$ in (7.7-3), multiply by $P_m(y)$, and integrate using Exercise 7.8-2.

7.8-8 Choose $f(x) = (y - x)^{-1}$ in (7.8-3) and use (7.1-11). Note that $[p_m(x) - p_m(y)] \cdot (y - x)^{-1}$ is a polynomial in y of degree $m - 1$.

7.8-9 Use Exercises 7.7-9 and 7.8-3.

7.8-10 Use Exercises 7.6-2 and 7.8-3.

7.9-1 Expand the left side in powers of x and t . Show that the double series converges absolutely and change the order of summation.

7.9-2 Proceed as in Exercise 7.9-1.

7.9-3 Since both sides are polynomials in β , assume $\beta \in \mathbb{C}_>$. Use (5.9-19). Deform γ so that it encircles $[0, x]$ and use (5.11-4).

7.9-4 In (6.11-2) replace $a, c - a, x, y$ by $1, \beta, 1 - x/t, 1$. By (6.2-24) and the M test, the series converges uniformly on the circle $|t| = \rho$ if $\rho > |xz|/(1 - |z|)$. Multiply by e'/t and integrate around this circle using (5.11-4) and Exercise 7.9-3.

- 7.9-5 Expand in powers of t and change the order of summation.
 7.9-6 Multiply the equation in Exercise 7.9-1 by $(1-t)^{-2}$.
 7.9-7 Multiply two generating functions of the type in Exercise 7.9-1.
 7.9-8 Differentiate the equation in Exercise 7.9-1.
 7.9-9 Use Exercise 7.9-8, (7.9-3), and (4.4-8).
 7.10-1 Use Exercise 3.4-4.
 7.10-2 Expand the integrand by the binomial theorem and use Exercise 7.10-1. Simplify by (2.2-12), (2.2-7), and (7.10-4).
 7.10-3 Use (6.4-2), Theorem 6.9-1, and (3.2-1). Simplify as in Exercise 7.10-2.
 7.10-4 Write $2xt - t^2 = x^2 - (x-t)^2$. Set $f(x) = \exp(-x^2)$ and evaluate the Taylor coefficients of $f(x-t)$ at $t=0$ by (7.10-6).
 7.10-5 Use (6.10-12) and proceed as in the proof of Theorem 7.10-1.
 7.10-6 In Exercise 7.10-4 replace (x, t) first by $(x, t \cos \theta)$ and second by $(y, t \sin \theta)$, multiply the two series, and take the coefficient of t^n .

Chapter 8

- 8.1-1 Set $t = s^{1/2}$ and use (8.1-2).
 8.1-2 Set $\sin \theta = t^{1/2}$ and use (8.1-2).
 8.1-3 Set $\sin \theta = t^{1/2}$ and use (8.1-2).
 8.1-4 We prove the first form of the result, with which the second is connected by (6.10-5). Since the integrand is unchanged if p or q is replaced by its complex conjugate, assume p and q are in the upper half-plane. The integrand is the restriction to the real line of $[(t-p)(q-t)(t-\bar{p})(\bar{q}-t)]^{-1/2}$, which is holomorphic on the t plane cut along the line segments $[p, q]$ and $[\bar{p}, \bar{q}]$. Close the path of integration by a large semicircle in the upper half-plane, squeeze the contour down to the cut along $[p, q]$, and evaluate by (8.1-2). Note that $(p-\bar{p})(\bar{q}-q) = d_+^2 - d_-^2$.
 8.1-5 Choose t^2 as a new variable and use (8.1-2).
 8.1-6 With b fixed use Appendix B.4 to show that the integral is entire in a . Since R_{-a} is entire in a by Theorem 6.8-2, it suffices to assume $a, a' \in \mathbb{C}_+$. Represent the integrand by (6.6-6), use Appendix B.2 to change the order of integration, and integrate by (6.6-6) and (8.1-2). Use (6.8-15).
 8.2-1 The interior of a rectangle which passes through 0 but does not have a vertex there.
 8.2-2 The interior of a closed convex polygon with k vertices and k equal angles. The polygon passes through 0 in each case, but 0 is a vertex only in the first case.
 8.2-3 See Exercise 8.2-2 and use the Schwarz reflection principle to continue $z(u)$ analytically.
 8.3-1 In (6.10-20) let $x \rightarrow 0$ and use (8.3-4).
 8.3-2 Set $t = \alpha - \beta - \frac{1}{2}$ in Exercise 8.3-1 and use (5.9-21).
 8.3-3 Use (8.3-7) and Exercise 8.3-1.
 8.3-4 Use (8.3-7), Exercise 8.3-2, and (3.7-2).
 8.3-5 In (8.3-8) set $\alpha = 1 - y$ and $\beta = \gamma - 1 = x$.
 8.3-6 Evaluate $M = \sqrt{3} u(x_1)$ by (8.3-2) and (3.9-3).
 8.3-7 Show that $(dr/ds)^2 = 1 - r^4$. Use Exercise 8.1-5, (8.2-6), and (8.3-17). Compare with (8.2-7) and (8.2-10).
 8.3-8 If $0 < x < c$, $t = \int_0^x (c^p - v^p)^{-1/p} dv$. Set $v = xu^{1/p}$ and use (5.9-1). Consider the phase plane (points with coordinates x and $y = dx/dt$) to see that the time for x to increase from 0 to c is a quarter period if it is finite. Use (8.3-2) and (3.9-3).

8.3-9 By the permanence of functional relations, (6.9-7) remains valid if the arguments on both sides remain in $\mathbb{C}_0$. They will do so if we replace (x, y) by $(x + \varepsilon, -x + \varepsilon)$, where $x, -x \in \mathbb{C}_0$ and $\varepsilon > 0$. Let $\varepsilon \rightarrow 0$ and use (8.3-4).

8.3-10 Show continuity by proceeding as in the proof of Corollary 8.3-4. That proof shows that the ${}_2F_1$ series converges uniformly on $0 \leq u \leq 1$ if $|x| \leq 1$. Integrate term by term applying (4.4-8).

8.3-11 Set $x = 1$ in Exercise 8.3-10 and use Exercise 5.9-2, (4.4-1), and again Exercise 8.3-10. In Exercise 5.9-2 the restrictions on the parameters can be replaced by the condition $\gamma \in U$ because of (5.9-12), Corollary 6.3-4, and the permanence of functional relations.

8.3-12 Iterate the transformation in Exercise 8.3-11.

8.4-1 Since E_n is homogeneous of degree n , $\sum z_i D_i E_n = nE_n$.

8.4-2 By the permanence of functional relations we may assume $a, a' - M \in \mathbb{C}_>$. In (8.4-3) replace b by $b - m$ and expand $\prod (1 + xz_i)^{m_i}$ by (6.6-1).

8.5-1 The second expression follows from the first by rationalizing the denominator in the logarithm. By a similar procedure it is easy to show that the signs of $(z - x)^{1/2}$ and $(z - y)^{1/2}$ are immaterial. Since the R function is holomorphic in (x, y, z) on $\mathbb{C}_0^3$, it suffices to prove the first equality when y and z are not in the closed convex hull of x and the negative real axis. Use (8.5-5) with $z_1 = x$ and (6.9-16).

8.5-2 Use (8.5-5) with $z_1 = w$.

Chapter 9

9.2-1 Write $R_K(x^2, y^2) = (2/\pi x)R_F(0, 1, y^2/x^2)$ and use (9.2-10).

9.2-2 Interchange x and y in (9.2-12) and (9.2-13).

9.2-3 Set $t = s + \lambda$ and use (8.3-1) and (9.2-1).

9.3-1 Set $\sin^2 \theta = t$ and use (8.1-2) and Table 9.3-3. Set $\cos^2 \varphi = x/z$, $1 - k^2 \sin^2 \varphi = y/z$, $1 - \alpha^2 \sin^2 \varphi = \rho/z$.

9.3-2 Use (7.7-6), Table 9.3-2, (9.2-4), and Exercise 9.2-1.

9.3-3 Use (5.2-3), (5.9-8) with $i = 3$, and Table 9.3-1.

9.3-4 Apply (6.8-15) to the second term on the left. Use homogeneity and Table 9.3-4.

9.4-1 Integrate $ds = \beta^{-1}(y^2 + \beta^2)^{-1/2}[(\alpha^2 + \beta^2)y^2 + \beta^4]^{1/2} dy$ by Exercise 8.1-5.

9.4-2 In the notation of Example 9.4-2, $x^2 = l^2 r^2 = l^2(l^2/\alpha^2 + m^2/\beta^2 + n^2/\gamma^2)^{-1}$. Use Exercise 5.9-20, (6.8-15), and Exercise 4.3-6.

9.4-3 Represent the reciprocal distance by a Fourier integral and perform the integrations over $x, \dots, z'$. Use Exercise 5.9-20 to integrate over angles in the Fourier transform space. See *J. Math. Phys.* **2** (1961), 441 for details.

9.5-1 To obtain (6.10-1) set $z = 0$ and use (8.3-4) and (5.9-20). To obtain (6.10-20) set $z = y$ and use (5.2-3) and (5.9-20).

9.5-2 In (6.10-1) set $t = \beta = \frac{1}{2}$ and use Table 9.3-2 and (6.10-5).

9.5-3 Use the next to the last line of Table 9.3-2 to show $A(n) \rightarrow 0$. Use Exercise 9.5-2 to find $A(n) - A(n+1)$.

9.5-4 Use Exercise 9.5-3 and the solution of Exercise 6.10-2.

9.5-5 Set $t = -\frac{1}{2}$ in (9.5-5) and use Table 9.3-1 and (9.5-4).

9.5-6 See Carlson (1965a, Section 12).

9.6-1 Use Exercise 6.2-14, the solution of which remains valid if $z^s \in \mathbb{C}_0^k$.

9.6-2 Use (9.6-9) and similar equations for x_{n+1} and y_{n+1} .

9.7-1 Multiply (9.7-12) by λ^2 , subtract the equation with λ and μ interchanged, and use (9.7-11).

9.7-2 Set $e = x + y + z$, $f = xy + xz + yz$, $g = xyz$ in Exercise 9.7-1 and make the replacements specified in the Remark following Theorem 9.7-1.

9.7-3 Expand the square.

9.7-4 Multiply (9.7-12) by λ and subtract the equation with λ and μ interchanged.

9.7-5 Proceed as in solving Exercise 9.7-2 but start from Exercise 9.7-4.

9.8-1 In Exercise 8.5-2 set $(w, x, y, z) = (A^2, B^2, C^2, D^2)$. In (9.7-12) and (9.7-14) put $\lambda = 1$, $e = x + y + z$, $f = xy + xz + yz$, $g = xyz$ and then $x = A^2/(B^2 - A^2)$, $y = A^2/(C^2 - A^2)$, $z = A^2/(D^2 - A^2)$. In the second part eliminate $R_{-1}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ between (9.8-4) and Exercise 8.5-2.

9.8-2 Use (9.4-14) and (9.8-4).

9.8-3 Set $t = s^{1/2}$ and use (9.8-12).

9.8-4 In (9.8-12) let $x \rightarrow \infty$ and set $y = r$, $\alpha = \beta = \gamma = \delta = 1$, $a = -p$, $b = -\bar{p}$, $c = -q$, $d = -\bar{q}$. In the notation of Exercise 8.1-4, $U = d_{>}$, $V^2 - U^2 = -\delta^2$, $W^2 - U^2 = -d^2$.

9.8-5 Set $A = x$, $B = y$, $C = D = z$ in (9.8-2) and (9.8-4). Use (5.2-3) and (8.2-13). For the second part use (6.9-17) and (6.9-12).

9.8-6 Multiply both integrals by $a^{1/2}$ and choose t/a as a new variable of integration so that the polynomials on both sides become monic. They can now be identified with the polynomials in (9.8-3), and coefficients can be compared.

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For extensive formularies see Abramowitz (1964), Erdélyi (1953), Gradshteyn (1965), Hansen (1975), Luke (1969, 1975), Magnus (1966), Oberhettinger (1973a), and Tölke (1950). For tables of integrals and integral transforms see Gradshteyn (1965), Erdélyi (1954), Luke (1962), Byrd (1971), Tölke (1950), Colombo (1972), and Oberhettinger (1972, 1973b, c, 1974). For numerical tables see Abramowitz (1964), Jahnke (1960), Luke (1975), Fletcher (1962), Lebedev (1960), Burunova (1960), and the journal *Mathematics of Computation*. For computational methods see Gautschi (1975) and Luke (1975).

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